shear stress (τ)

σ

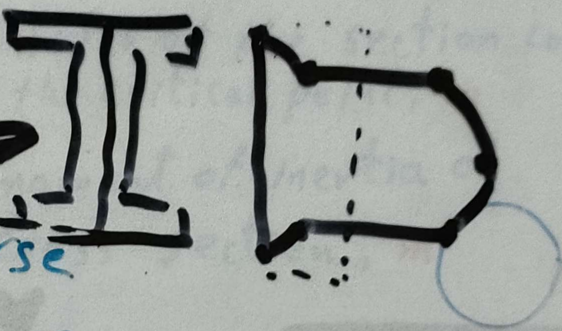
σ

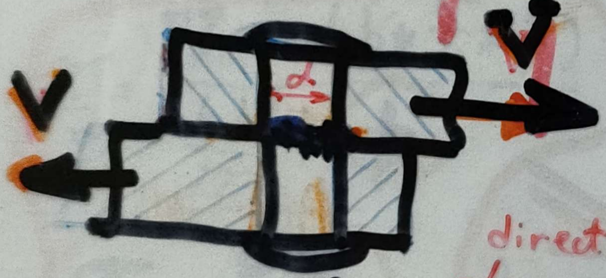
δ

τ

Shear force V — Torsion

direct shear — transverse shear



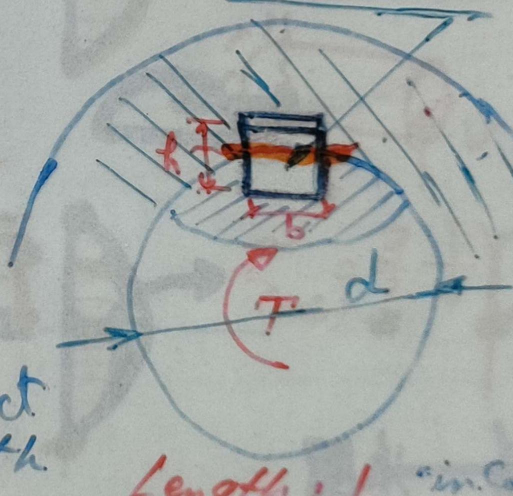


$$\tau = \frac{V}{A}$$

direct shear force

$A = \frac{\pi d^2}{4}$

Feather key



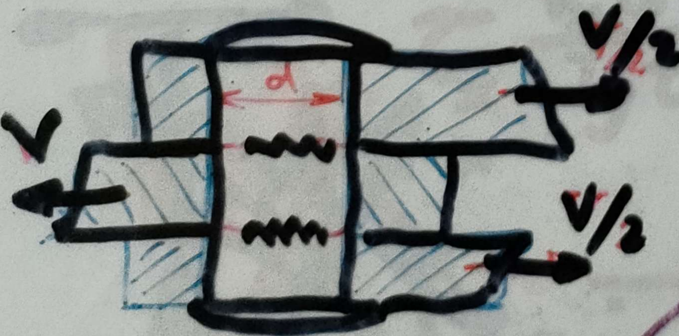
L: Contact Length

Length: L "in contact"

$$\tau = \frac{V}{A}$$

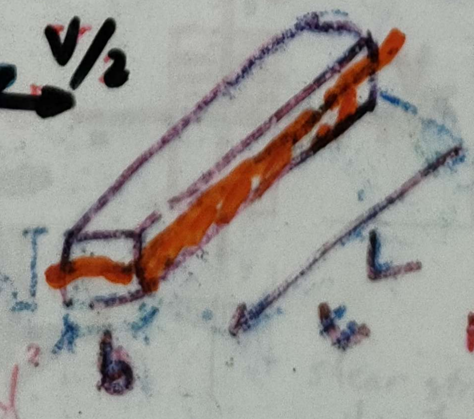
shear force

~~Contact~~ area "(contact length)"



$$\tau = \frac{V}{2A}$$

$A = \frac{\pi}{4} d^2$



$$V = \frac{T}{d/2}$$

$$A = b \cdot L$$

$$\tau = \frac{VQ}{It}$$

Transverse shear load on the cross section, N

moment of the cross-sectional area of the member, above or below the critical point, with respect to the neutral axis, m^3 (1^{st} moment of area) m^3

$$\boxed{Q = \int y\, dA}$$

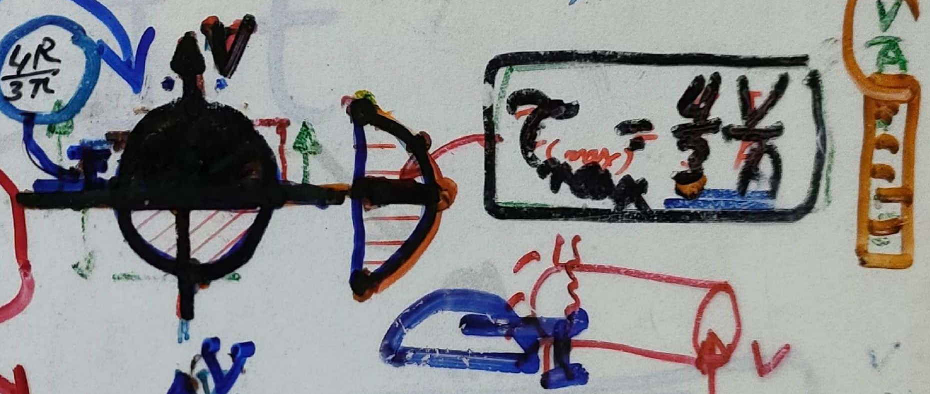
Transverse shear, Pa or N/m^2

width of the section containg the critical point, m

moment of inertia of cross section, m^4

$$Q = \left(\frac{\pi d^2}{8}\right) \cdot \frac{4d}{6\pi}$$

$$I = \frac{\pi d^4}{64} \qquad t = d$$



$$\boxed{\tau_{(max)} = \frac{4}{3}\frac{V}{A}}$$

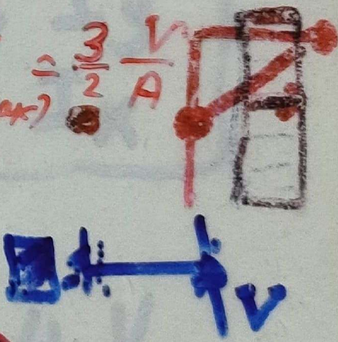
$$\frac{V}{A}$$

$$\tau_{\substack{max\\C.G}} = \frac{V * \left(b * \frac{h}{2} * \frac{h}{4}\right)}{\frac{bh^3}{12} * b}$$

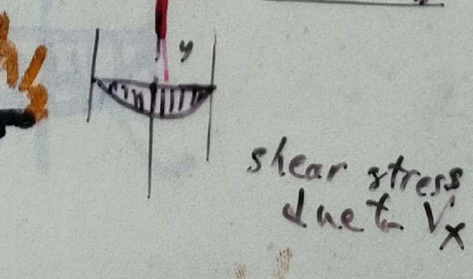
$$\tau_{CG} = \frac{3}{2}\frac{V}{A}$$

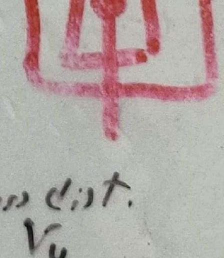
$$\boxed{A = b \cdot h}$$

$$\tau_{(max)} = \frac{3}{2}\frac{V}{A}$$



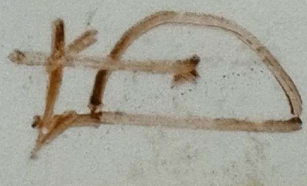
$$\tau_x = \frac{V_x Q}{I_y t}, \quad \tau_y = \frac{V_y Q}{I_x t}$$





shear stress dist. due to V_y

shear stress dist. due to V_x



Transverse shear

τ

Shear force [N]

1st. Moment of area. m^3 mm^3

MPa

$$\text{Pa} \leftarrow \tau = \frac{VQ}{It}$$

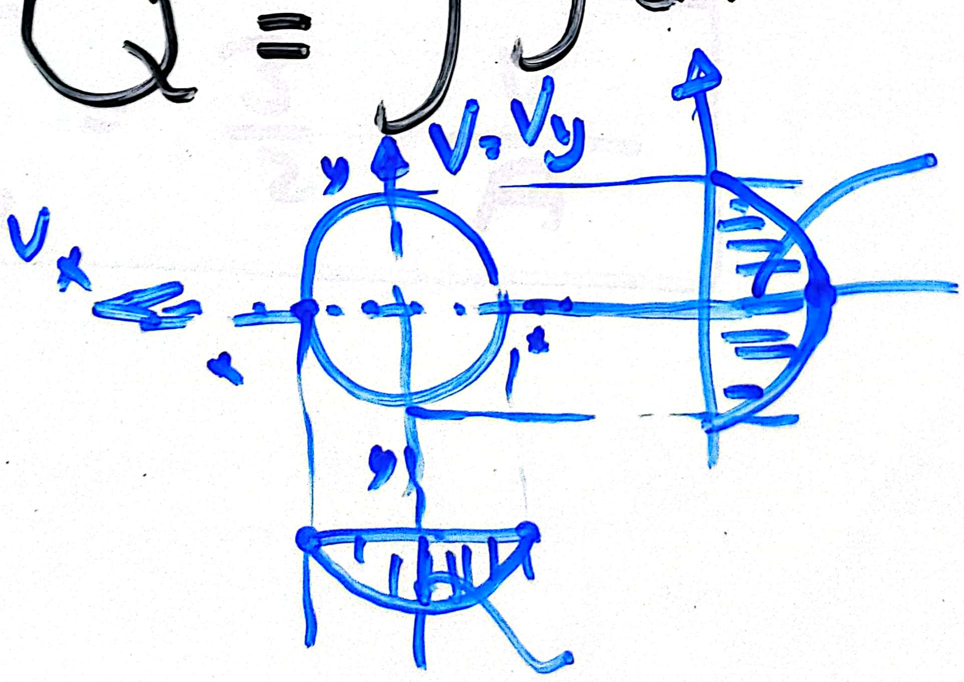
m^4 mm^4 Moment of Inertia

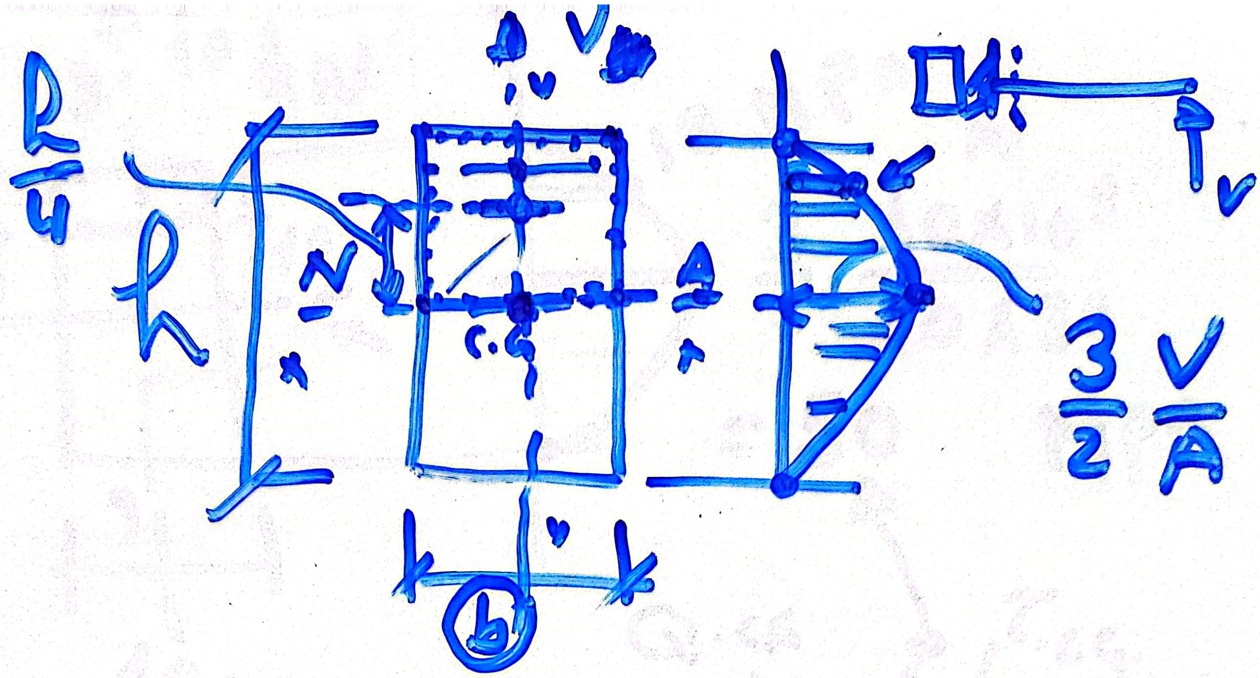
m mm

$$\boxed{\begin{array}{l} V_x \Rightarrow I_y \\ V_y \Rightarrow I_x \end{array}}$$

$$Q = \int y\,dA$$

$\frac{4}{3}\frac{V}{A}$

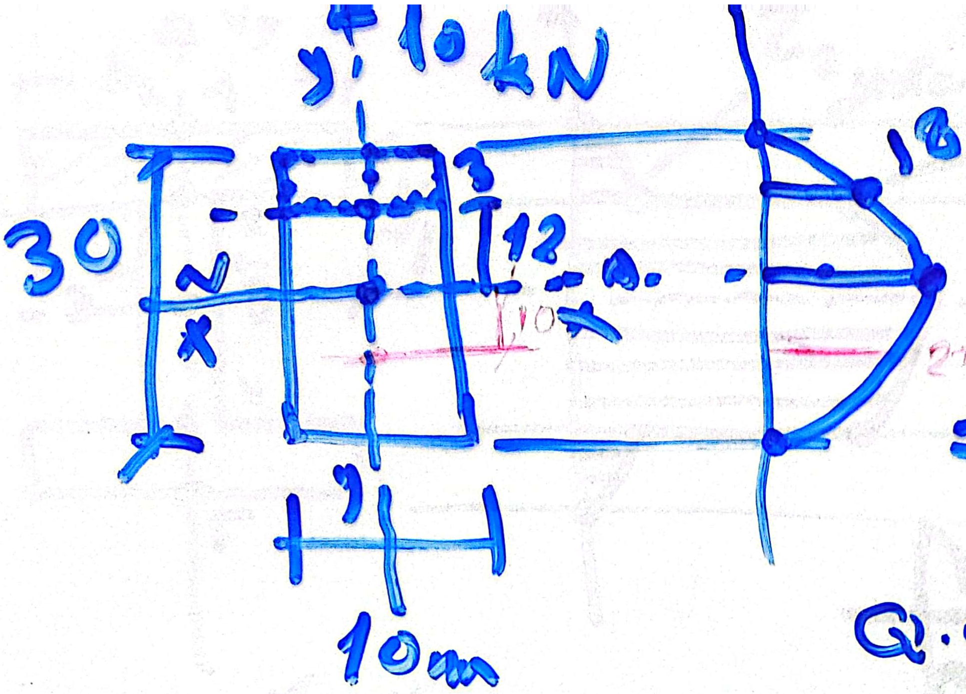




$$\tau = \frac{V_y}{I_x} \cdot \frac{Q}{t}$$

$$= \frac{V \cdot \left[b \cdot \frac{h}{2} \cdot \frac{h}{4} \right]}{\frac{b \cdot h^3}{12} \cdot b}$$

$$\boxed{\tau_{C.G} = \frac{3}{2} \cdot \frac{V}{A}}$$



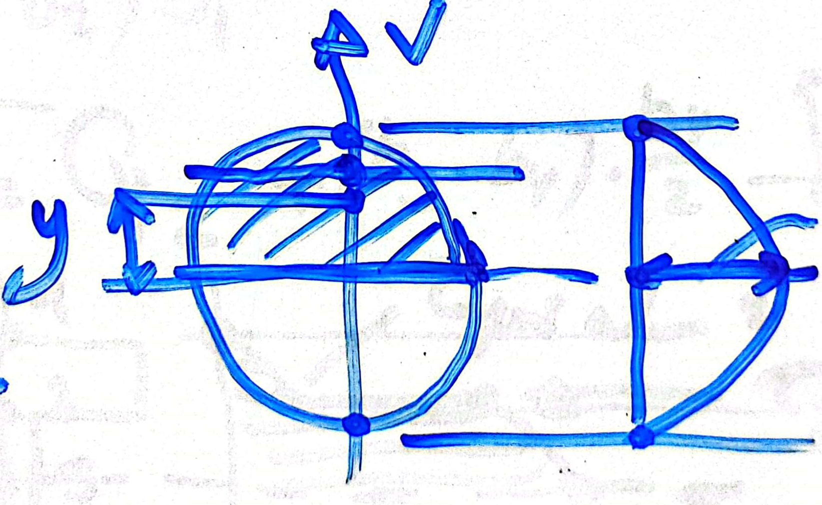
$$\frac{3}{2}\cdot\frac{\overset{N}{10\times10^3}}{10\times30} = 50 \text{ Mpa}$$

Q.c.g τ.c.g.

$$\tau = \frac{\overset{N}{10\times10^3} \times \left(\overset{30\times10\times13.5}{\cancel{3\times10\times13.5}}\right)}{\frac{10(30)^3}{12}\cdot 10}$$

$Q_{w\,12\,mm}$

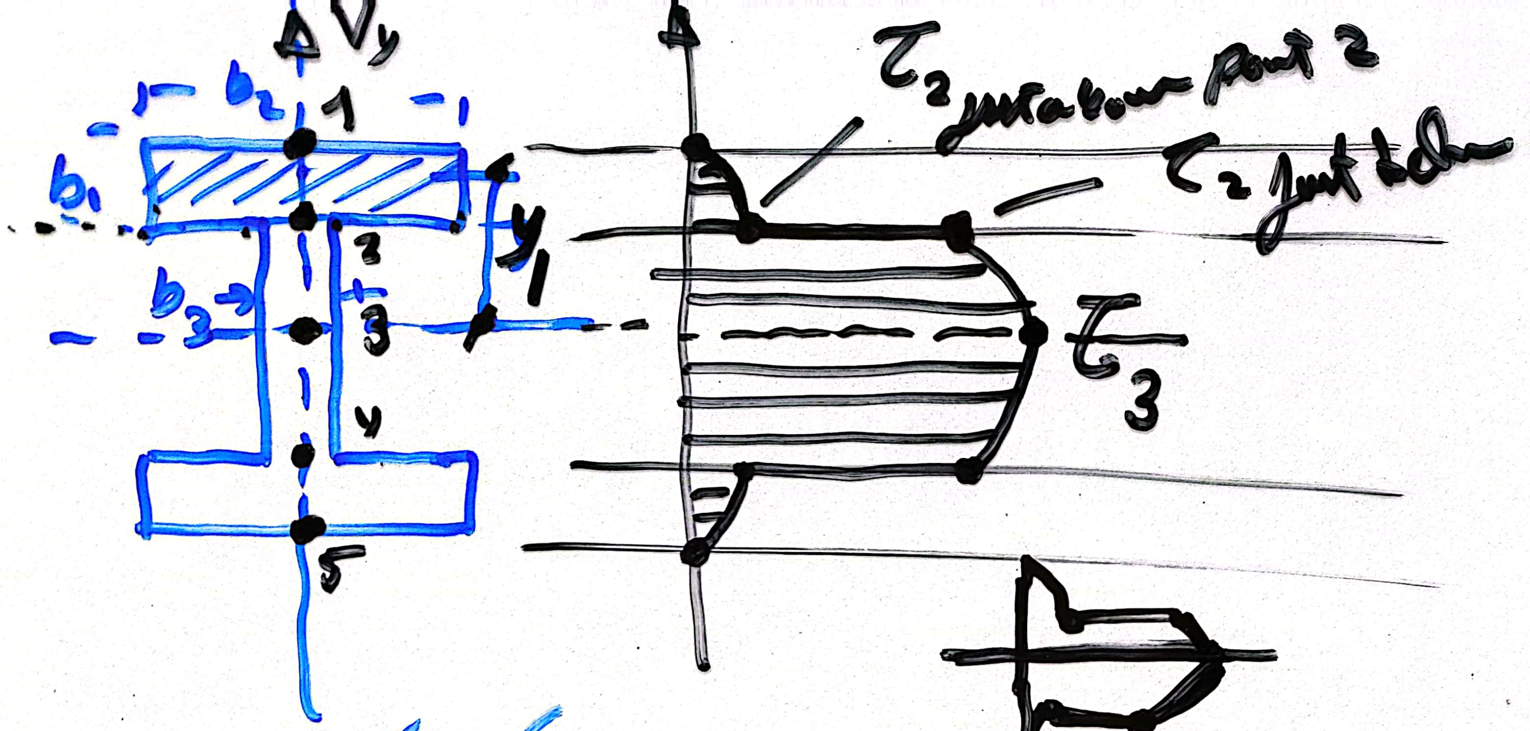
$= 18$ Mpa



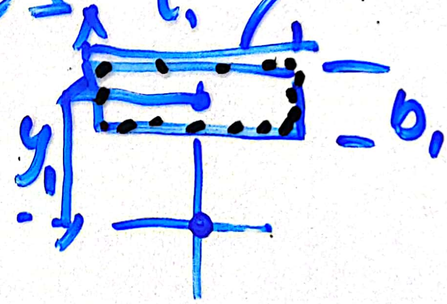
$$\frac{4R}{3\pi}$$

$$\frac{4}{3}\frac{V}{A}$$

$$\tau = \frac{V\left(\frac{1}{2}\frac{\pi}{4}d^2\cdot y\right)}{\frac{\pi d^4}{64}\cdot d} \dots$$



$$\tau = \frac{V_y\, Q}{I\; t}$$



t_2 < b_2 just above 2
b_3 just below 2

$Q_2 = (b_1 \times b_2)\, y_1$

$$Q'_3 = Q_2 + (b_3 \cdot b_4) \cdot \frac{b_4}{2}$$